

Learning rate experiments

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Batching and epochs

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Updating parameters

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Loss functions

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Explain a crossing for Learning rate experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Learning rate experiments. Proposed explanation: “Crossing the minimum means divergence”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Compare a new objective independently

Use $J(a) = (a - 1)^2$ with gradient $2(a - 1)$, start $a = 0$ and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace slow progress

For rate 0.05, start $v = 0$ and compute two updates and their training MSE values using the lesson objective.

My reasoning and evidence

6. W2 Explain a crossing

At rate 0.3 the first update gives $v = 3$, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved divergence?

My reasoning and evidence

7. W3 Preserve an update after rescaling loss

Replace MSE by the sum of squared errors on the two training pairs. Which rate reproduces the MSE run at rate 0.3? Explain using gradients.

My reasoning and evidence

8. W4 Compare a new objective independently

Use $J(a) = (a - 1)^2$ with gradient $2(a - 1)$, start $a = 0$ and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.

My reasoning and evidence

9. W5 Find an unfair comparison

Run A starts $v = 0$ with full-data batches for four updates. Run B starts at the final A value, changes both rate and batch size, then compares final training losses. Can the difference be attributed only to rate? Propose a repair.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled learning rate experiments evidence sheet to Image-pattern Network Investigation. Show how you can run a controlled comparison of learning rates. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Learning rate experiments

Compare learning rates with identical starting loss and update counts.

Loss $J(a) = (a-4)^2$; gradient $2(a-4)$; start $a = 0$. Compare two updates at rates 0.25 and 0.75.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Learning rate experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Rate0.25

First $a = 2$, loss 4; second $a = 3$, loss 1.



2. Rate0.75

First $a = 6$, loss 4; second $a = 3$, loss 1.



3. Inspect paths

Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

CHANGE ONE THING

Add a rate 1 run from $a = 0$.

Explain the changed result using the matching lesson.

What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare learning rates with identical starting loss and update counts.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

1. Inspect paths
2. Rate0.25
3. Rate0.75

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Inspect paths Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it. Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back. Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

Practice 2: explain the deciding evidence

Changed-case practice: Add a rate 1 run from $a = 0$. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Its path is 8 then 0, with loss 16 at both steps.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Its path is 8 then 0, with loss 16 at both steps. The update alternates across 4 without reducing the initial loss.

Repair a missing building block

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Updating parameters (ai-5/11). Goal: Apply a learning rate to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
